

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("avg-household-size.csv")

print(df.head())
print(df.info())
print(df.describe())

```

```

   statefips  countyfips  avghouseholdsize  geography
0         2         13         2.43  Aleutians East Borough, Alaska
1         2         16         3.59  Aleutians West Census Area, Alaska
2         2         20         2.77  Anchorage Municipality, Alaska
3         2         50         3.86  Bethel Census Area, Alaska
4         2         60         2.50  Bristol Bay Borough, Alaska

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 3220 entries, 0 to 3219
```

```
Data columns (total 4 columns):
```

```

#   Column          Non-Null Count  Dtype
---  -
0   statefips      3220 non-null    int64
1   countyfips     3220 non-null    int64
2   avghouseholdsize 3220 non-null    float64
3   geography      3220 non-null    object

```

```
dtypes: float64(1), int64(2), object(1)
```

```
memory usage: 100.8+ KB
```

```
None
```

```

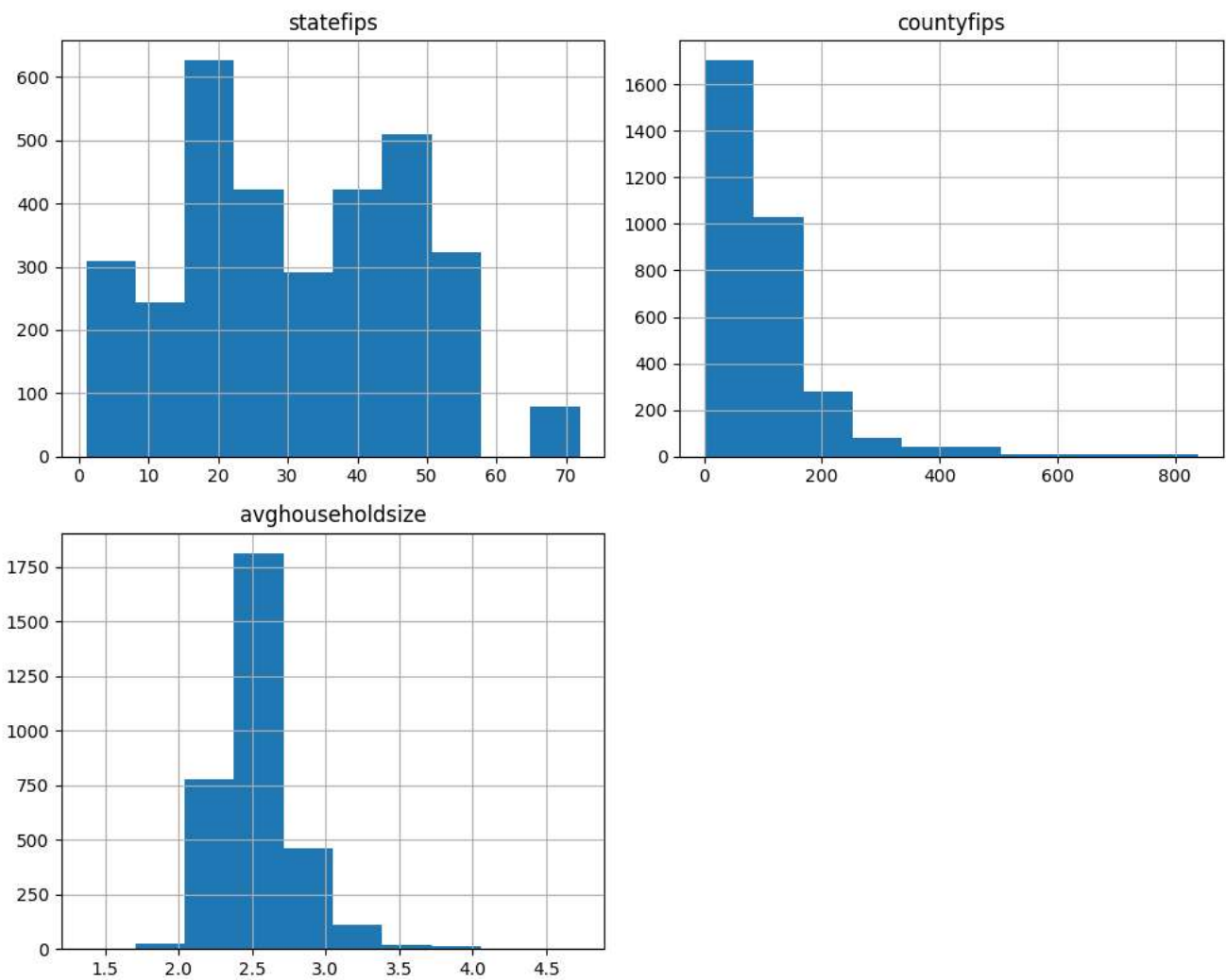
   statefips  countyfips  avghouseholdsize
count  3220.000000  3220.000000  3220.000000
mean    31.290683   102.922050    2.540755
std     16.277207    106.693846    0.267654
min      1.000000     1.000000    1.370000
25%     19.000000    35.000000    2.380000
50%     30.000000    79.000000    2.510000
75%     46.000000   133.000000    2.650000
max      72.000000   840.000000    4.730000

```

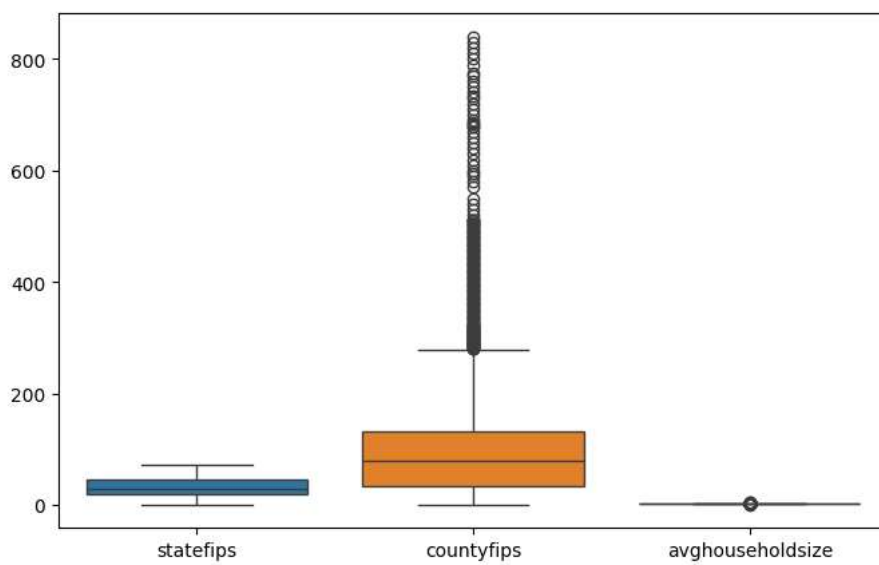
```

df.hist(figsize=(10,8))
plt.tight_layout()
plt.show()

```



```
plt.figure(figsize=(8,5))
sns.boxplot(data=df.select_dtypes(include=np.number))
plt.show()
```



```
print("First 5 rows:")
print(df.head())
```

```
First 5 rows:
statefips  countyfips  avghouseholdsize  geography
```

0	2	13	2.43	Aleutians East Borough, Alaska
1	2	16	3.59	Aleutians West Census Area, Alaska
2	2	20	2.77	Anchorage Municipality, Alaska
3	2	50	3.86	Bethel Census Area, Alaska
4	2	60	2.50	Bristol Bay Borough, Alaska

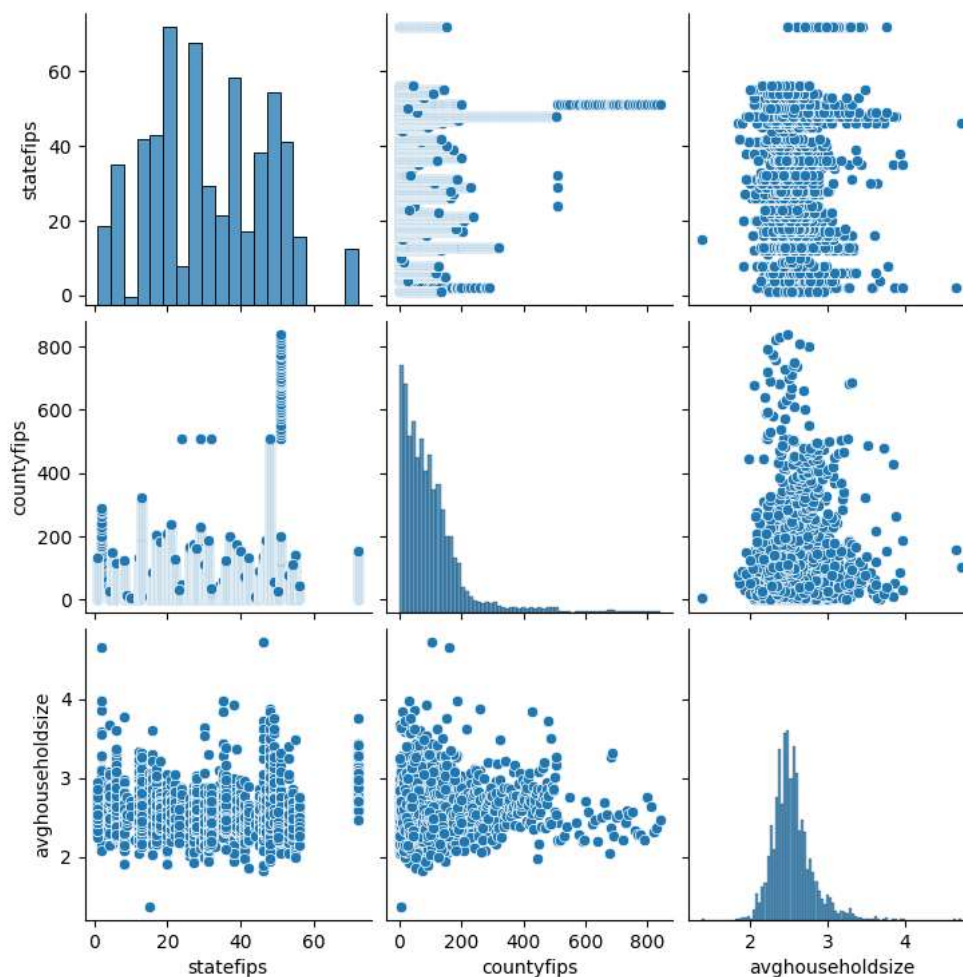
```
print("\nColumns:")
print(df.columns)
```

```
Columns:
Index(['statefips', 'countyfips', 'avghouseholdsize', 'geography'], dtype='object')
```

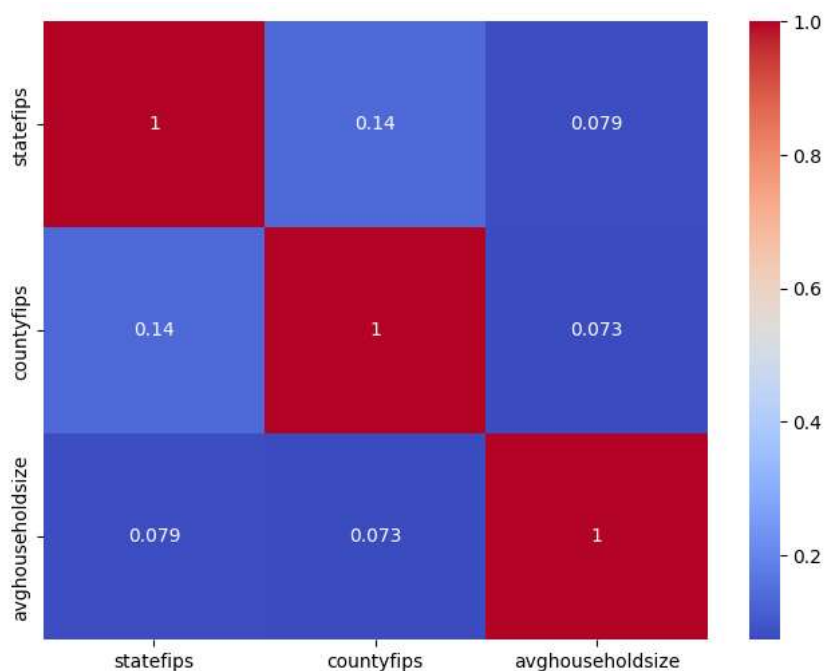
```
print("\nInfo:")
print(df.info())
```

```
Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3220 entries, 0 to 3219
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   statefips       3220 non-null   int64
1   countyfips      3220 non-null   int64
2   avghouseholdsize 3220 non-null   float64
3   geography       3220 non-null   object
dtypes: float64(1), int64(2), object(1)
memory usage: 100.8+ KB
None
```

```
sns.pairplot(df)
plt.show()
```



```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()
```



```
if len(df.select_dtypes(include='object').columns) >= 2:
    cols = df.select_dtypes(include='object').columns
    print(pd.crosstab(df[cols[0]], df[cols[1]]))
```

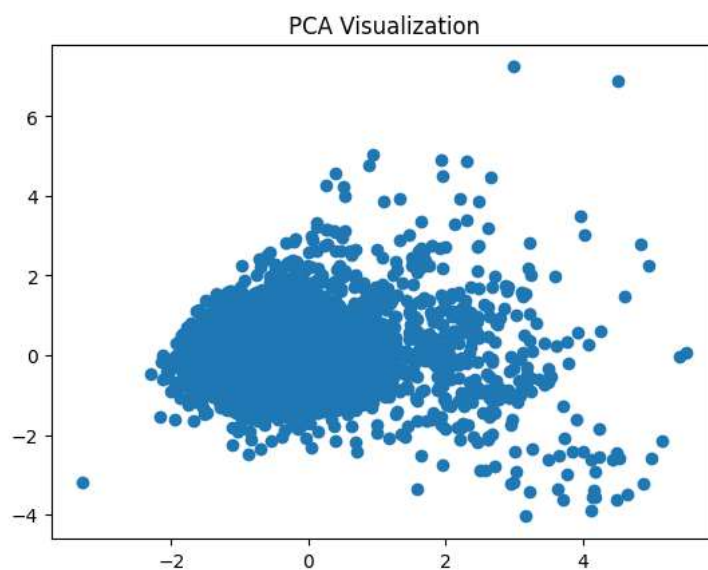
```
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

num_df = df.select_dtypes(include=np.number).dropna()

scaler = StandardScaler()
scaled = scaler.fit_transform(num_df)

pca = PCA(n_components=2)
pca_result = pca.fit_transform(scaled)

plt.scatter(pca_result[:,0], pca_result[:,1])
plt.title("PCA Visualization")
plt.show()
```



```
print(df.isnull().sum())
```

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
statefips      0
countyfips     0
avghouseholdsize  0
geography      0
dtype: int64
```

```
le = LabelEncoder()
for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col])
```

```
print("Mean:\n", df.mean())
print("Median:\n", df.median())
print("Std Dev:\n", df.std())
```

```
Mean:
statefips      31.290683
countyfips     102.922050
avghouseholdsize  2.540755
geography     1609.500000
dtype: float64
Median:
statefips      30.00
countyfips     79.00
avghouseholdsize  2.51
geography     1609.50
dtype: float64
Std Dev:
statefips      16.277207
countyfips     106.693846
avghouseholdsize  0.267654
geography     929.678260
dtype: float64
```

```
cat_cols = df.select_dtypes(include='object').columns
```

```
if len(cat_cols) > 0:
    df[cat_cols[0]].value_counts().plot.pie(autopct='%1.1f%%')
    plt.title(cat_cols[0])
    plt.show()
```

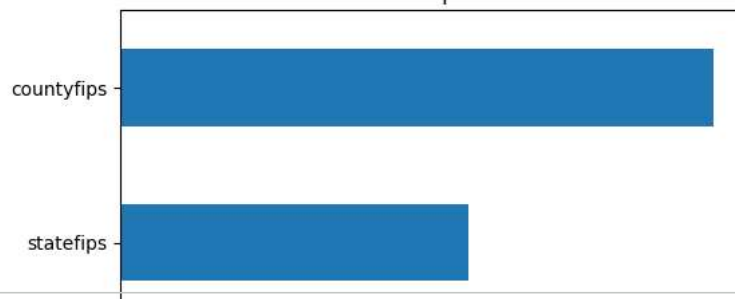
```
from sklearn.metrics import r2_score, mean_squared_error
X = df[["statefips", "countyfips"]]
y = df["avghouseholdsize"]
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
MSE: 0.06984505826086954
RMSE: 0.26428215653136616
R2 Score: 0.04840013860441539
```

```
importances = pd.Series(model.feature_importances_, index=X.columns).sort_values()
importances.plot(kind='barh', figsize=(6, 3), title="Feature Importance")
plt.show()
```

Feature Importance



```
plt.figure(figsize=(6, 6))
plt.scatter(y_test, y_pred, alpha=0.5)
lims = [min(y_test.min(), y_pred.min()), max(y_test.max(), y_pred.max())]
plt.plot(lims, lims, 'r--')
plt.xlabel("Actual avghouseholdsize")
plt.ylabel("Predicted avghouseholdsize")
plt.title("Actual vs Predicted")
plt.show()
```

Actual vs Predicted

